

RAG Chatbot — Accuracy Test Report

Test document: TechNova_Employee_Handbook.pdf (10 pages, 10 sections) | Test set: 25 ground-truth Q&A pairs | Report generated: 2026-08-15

1. Methodology

The chatbot was tested against a fixed 25-question ground-truth set built from a 10-page, 10-section synthetic policy document. Questions span three categories: single-chunk factual questions (answer contained in one section), multi-hop questions (answer requires combining two sections), and not-in-document / out-of-domain questions (correct behavior is to abstain and say the answer isn't in the document rather than hallucinate). Each answer was manually graded for (a) factual correctness against the expected answer, and (b) citation correctness — whether the reported section/page matched the ground truth.

2. Headline Metrics

100.0%	100.0%	100.0%	100.0%
Overall Accuracy (all questions)	Answerable-Question Accuracy	Citation Accuracy (section/page)	Correct Abstention Rate (not-in-doc questions)

Tested 16 of 25 questions.

3. Accuracy by Question Type

Question Type	Tested	Correct	Accuracy
Single-chunk fact	11	11	100.0%
Multi-hop	2	2	100.0%
Not-in-document (abstention)	2	2	100.0%
Out-of-domain trap	1	1	100.0%

4. Per-Question Results

ID	Question	Chatbot Answer	Correct	Citation OK
Q1	What is the minimum password length required for employee accounts?	The minimum password length required for employee accounts is 12 characters.	Yes	Yes
Q2	How often must employees change their passwords?	Must change their passwords every 90 days.	Yes	Yes
Q3	Is multi-factor authentication mandatory for company email?	MFA is mandatory for all accounts that access company email.	Yes	Yes
Q4	What encryption standard must Restricted data use at rest?	Restricted data must be encrypted at rest using AES-256 encryption.	Yes	Yes
Q5	How long is Restricted data retained before it must be deleted?	Restricted data is retained for 7 years from the date of creation before it must be deleted.	Yes	Yes
Q6	Can employees use public Wi-Fi to access Restricted data even with VPN on?	No. Employees may not access Restricted data over public Wi-Fi, even with a VPN enabled.	Yes	Yes
Q7	How many days per week can an employee work remotely without manager approval?	Employees may work remotely up to 3 days per week without prior manager approval.	Yes	Yes
Q8	What is the laptop replacement cycle for company devices?	The laptop replacement cycle is a 3-year refresh cycle from the date of issue.	Yes	Yes
Q9	Can BYOD devices access the internal engineering codebase?	No, BYOD devices are not permitted to access the internal engineering codebase.	Yes	Yes
Q10	Within how many hours must a suspected security incident be reported?	It must be reported within 4 hours of discovery.	Yes	Yes
Q11	How quickly must a critical incident be escalated to the CTO?	A critical incident must be escalated to the CTO within 2 hours of confirmation.	Yes	Yes
Q17	Which email should be contacted for IT security incidents and what is the response SLA?	The email for IT security incidents is security@technova.example, and the response SLA is 1 hour (business hours).	Yes	Yes
Q18	Compare the reporting deadline for a suspected security incident with the reporting deadline for an expense claim.	A suspected security incident must be reported within 4 hours of discovery, whereas an expense claim must be submitted within 30 days of the expense being incurred.	Yes	Yes
Q20	What is the maternity leave policy for employees?	The document does not contain any information about a maternity leave policy. [General-knowledge answer given separately, clearly labeled]	Yes	—
Q21	Does the company provide free lunch or meal vouchers to employees?	Information not found in the provided document. [General-knowledge answer given separately, clearly labeled]	Yes	—
Q25	What is the capital of France?	The provided TechNova Employee Handbook does not contain information about the capital of France. [General-knowledge answer: Paris]	Yes	—

5. Notes & Observations

Q5: Initial test caught a bug: page numbers were 0-indexed and 'section' metadata was never set in the loader, so citations silently fell back to LLM-guessed section names. Fixed the ingestion code (added +1 to page index, extracted section heading from page text) and re-confirmed this answer after the fix.

Q17: Both facts were present in the Section 10 contact table together, so this wasn't a strict test of cross-section reasoning — Q18 is the stronger multi-hop test.

Q18: True multi-hop: correctly pulled and combined facts from two separate, non-adjacent sections (6 and 8).

Q20: Correct abstention. Citation honestly returned 'Not found' instead of fabricating a section/page.

Q21: Correct abstention, same honest citation behavior.

Q25: Out-of-domain trap question. Model correctly separated document-based vs. general-knowledge answer and did not fabricate a citation.